



# Place & Manner

Dr. Josef Fruehwald

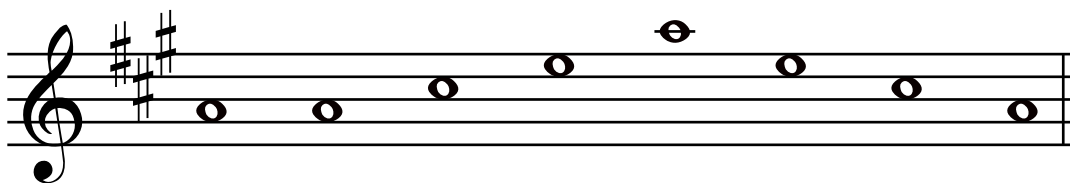
2026-09-03

## Reading

Zsiga Chapter 2

## Transcription

Transcription

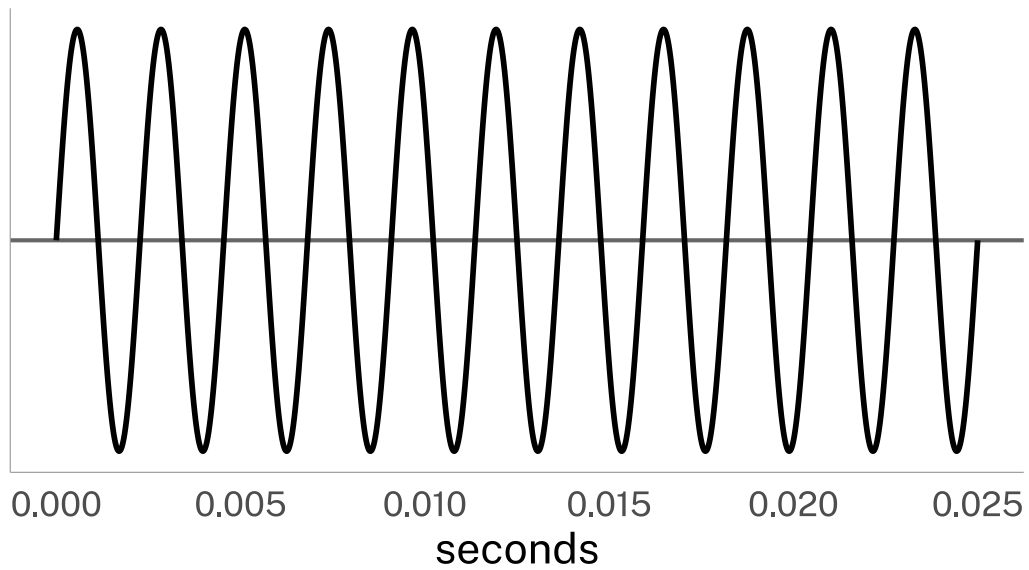


## Notation

But what is “A”?



440 Hertz



### Transcription

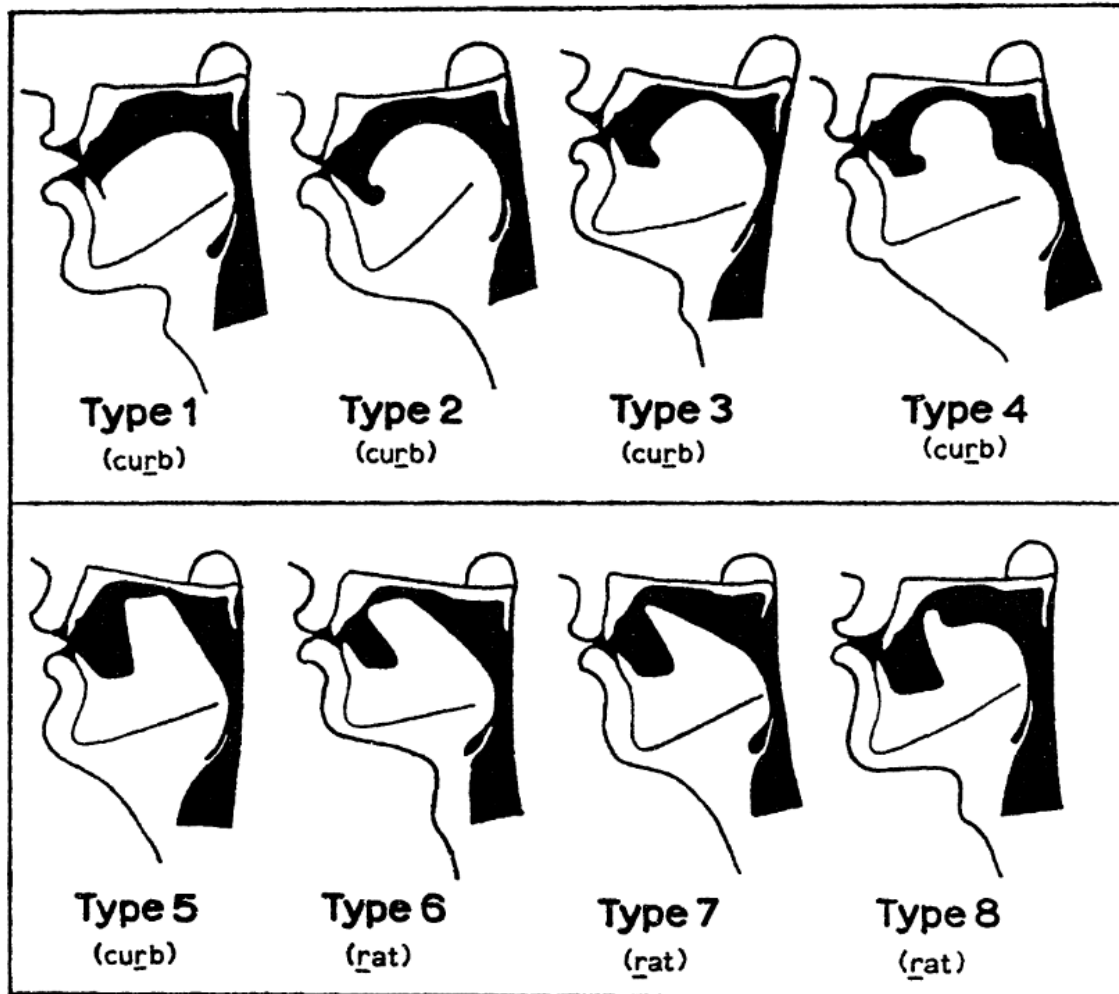
There have been different conventions for what frequency “Concert A” should be!

440Hz how?





ɪ how?



source: Delattre & Freeman (1968)



## The IPA!

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | ɹ        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## IPA Bits and Pieces

Symbol

Diacritics

d<sup>h</sup><sub>ɹ</sub>

## IPA Bits and Pieces

You *really* should not assume that the un-decorated symbol is more “basic” or “general” or “simpler” than the decorated symbol.

*diss* (to insult)

[dɪs]

*dis* (this)

[d̪ɪs]

## Place & Manner

### Place

- Where the point of greatest constriction is.



### Manner

- *What* happens at that place of constriction.

## Place

### Place

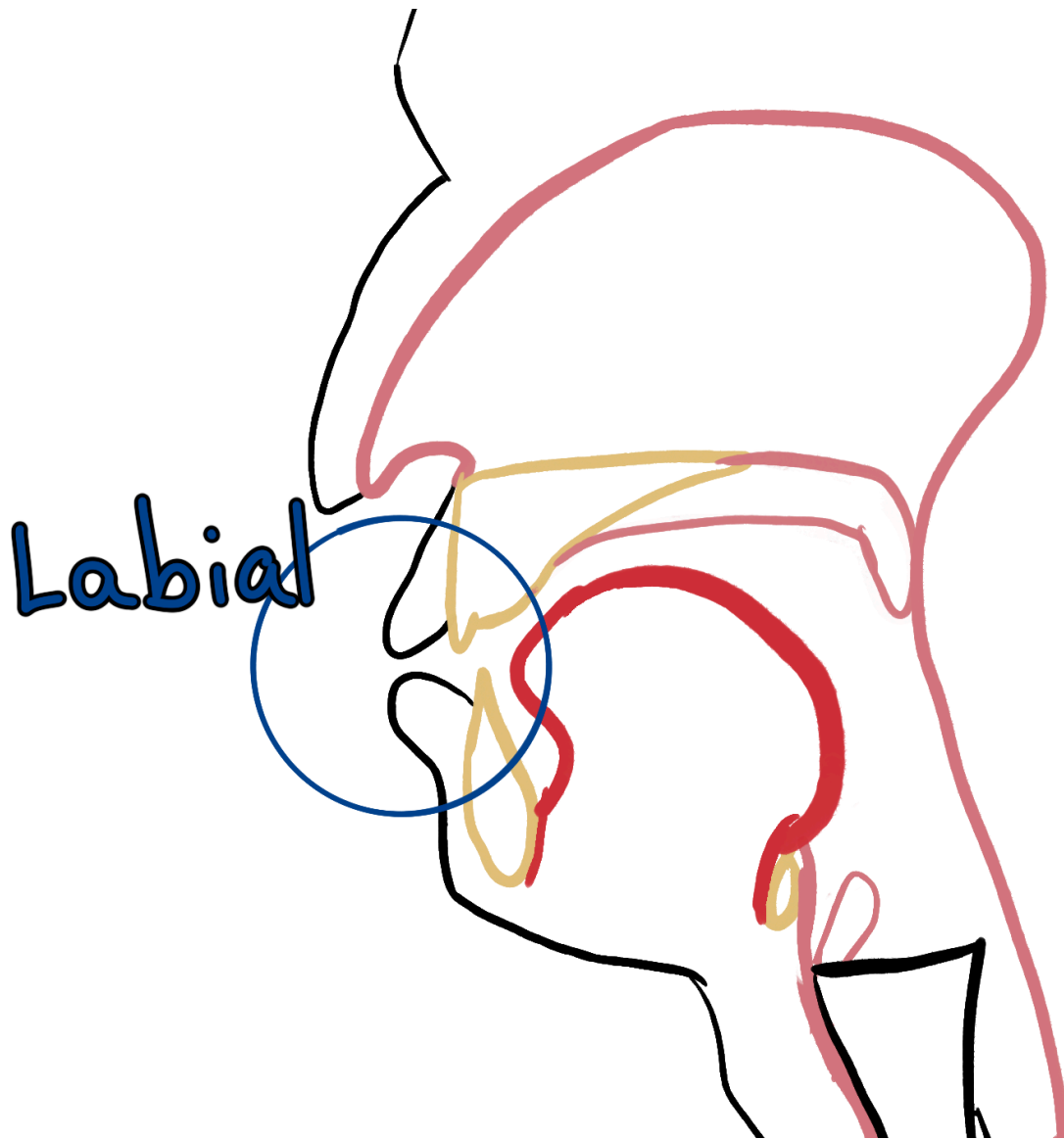
Most articulations have an *active* and a *passive* articulator

- Active: the articulator that moves
- Passive: the articulator that gets moved to

Most places of articulation are named for the location of the passive articulator.

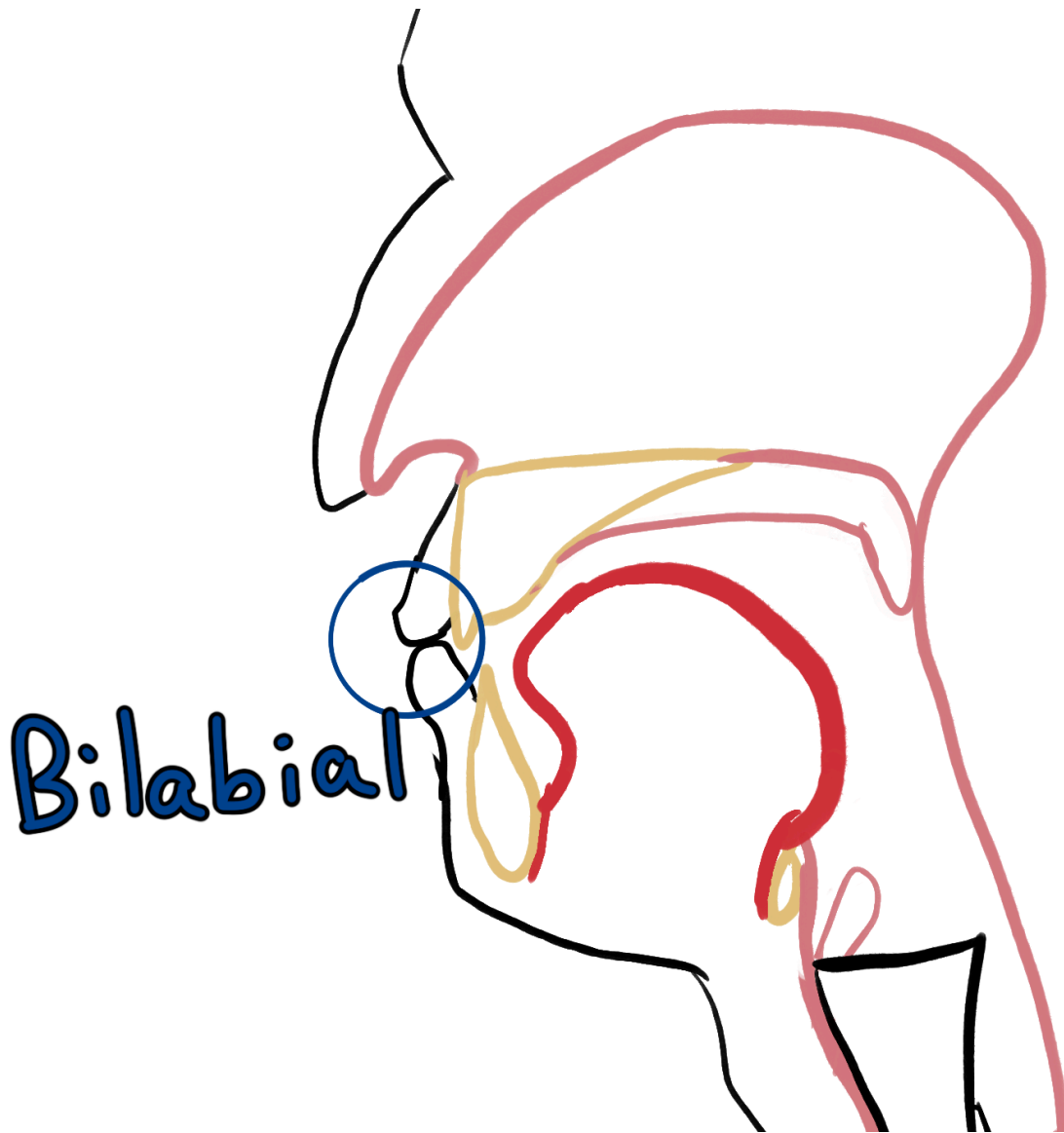


Place





Place





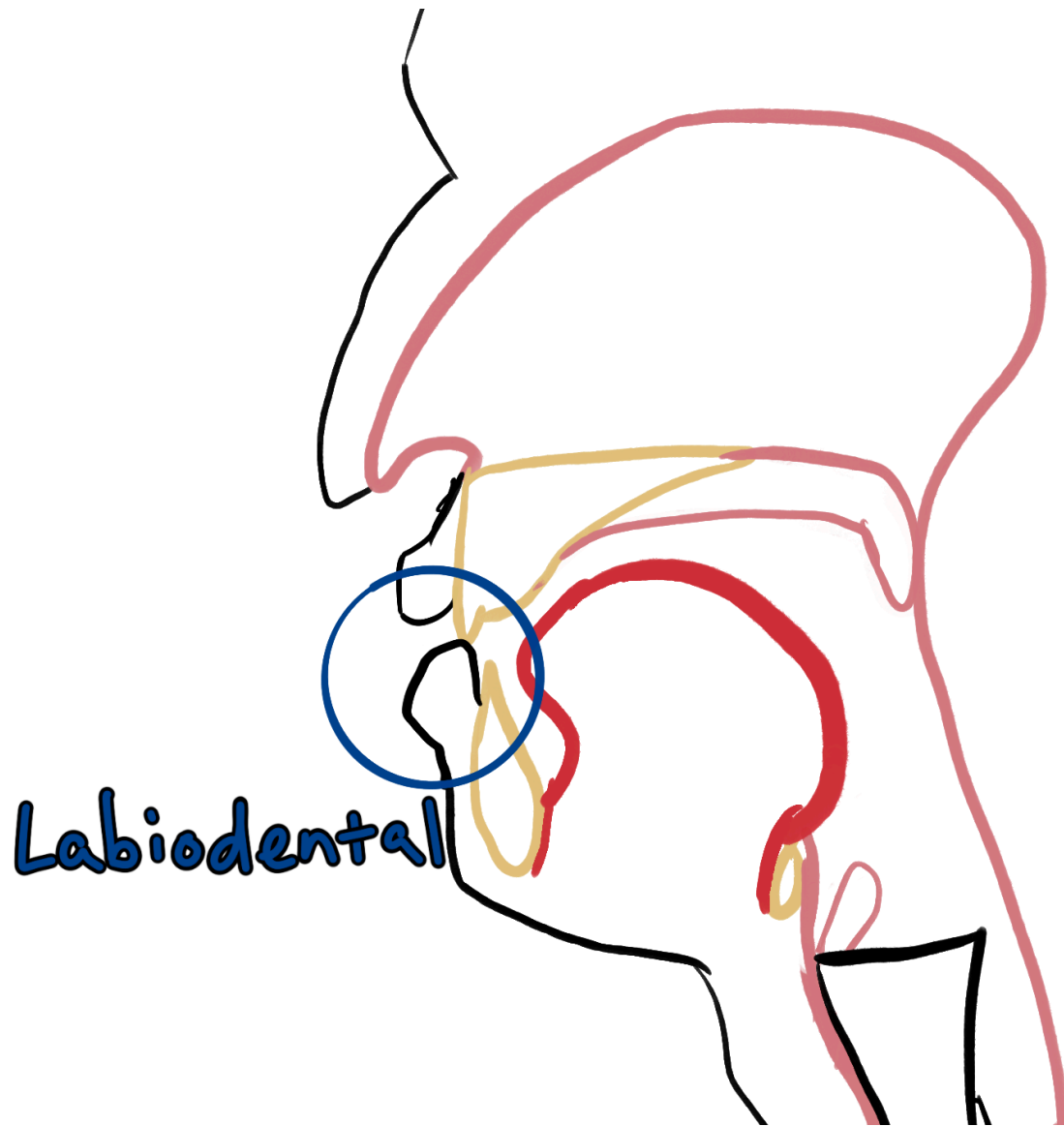
## Place

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |





Place



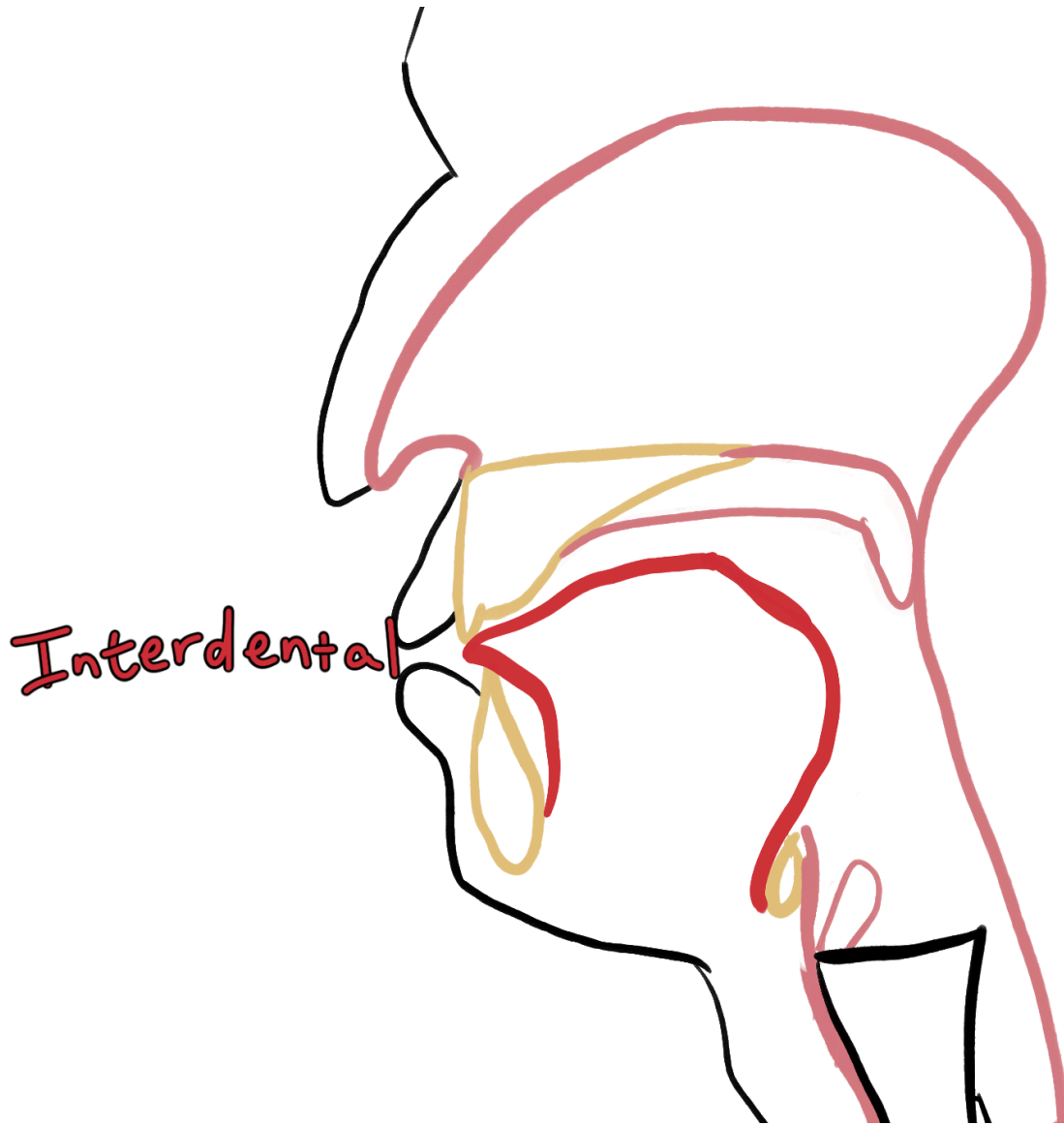


## Place

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
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| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
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| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |



Place



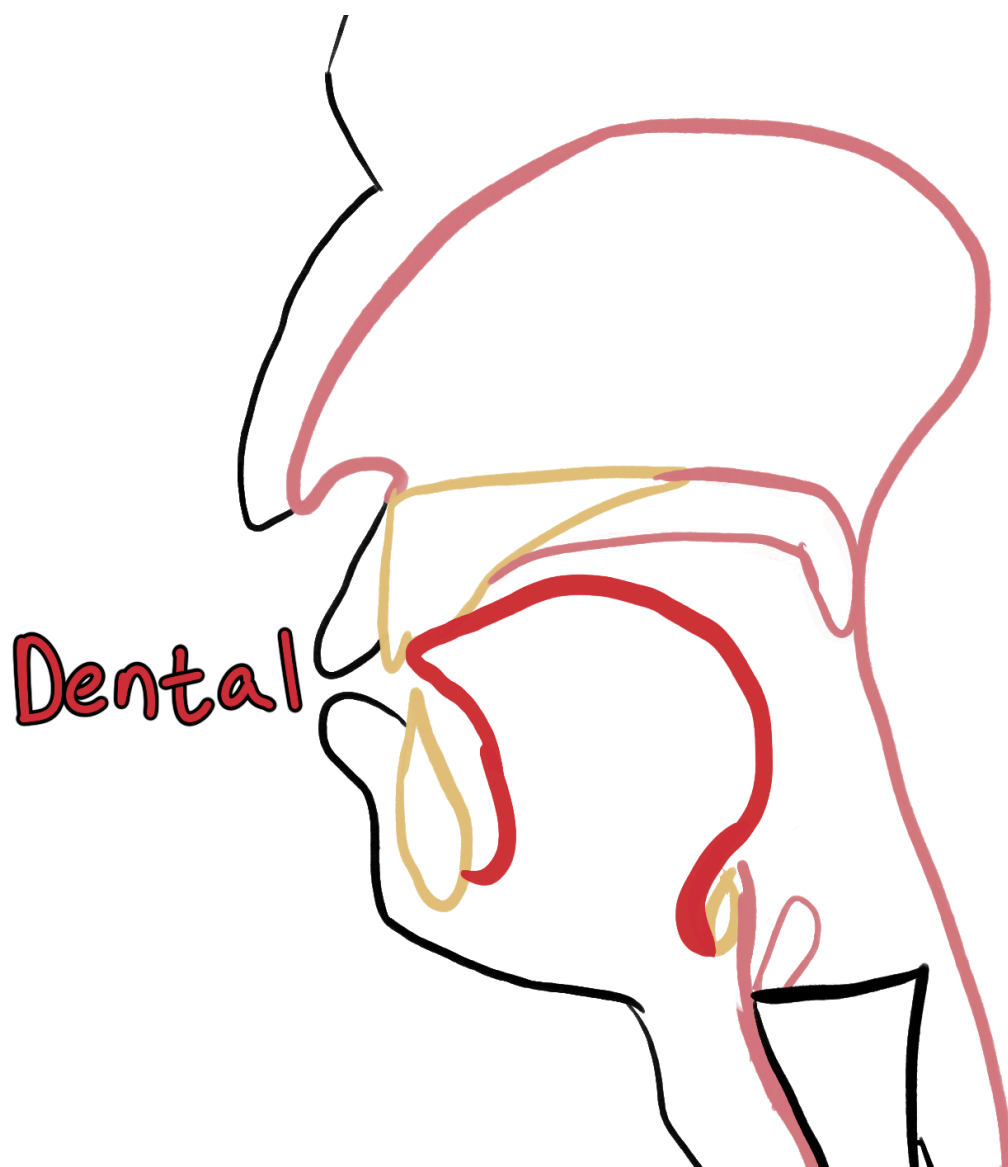


## Place

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
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| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |



Place



Place

Dental

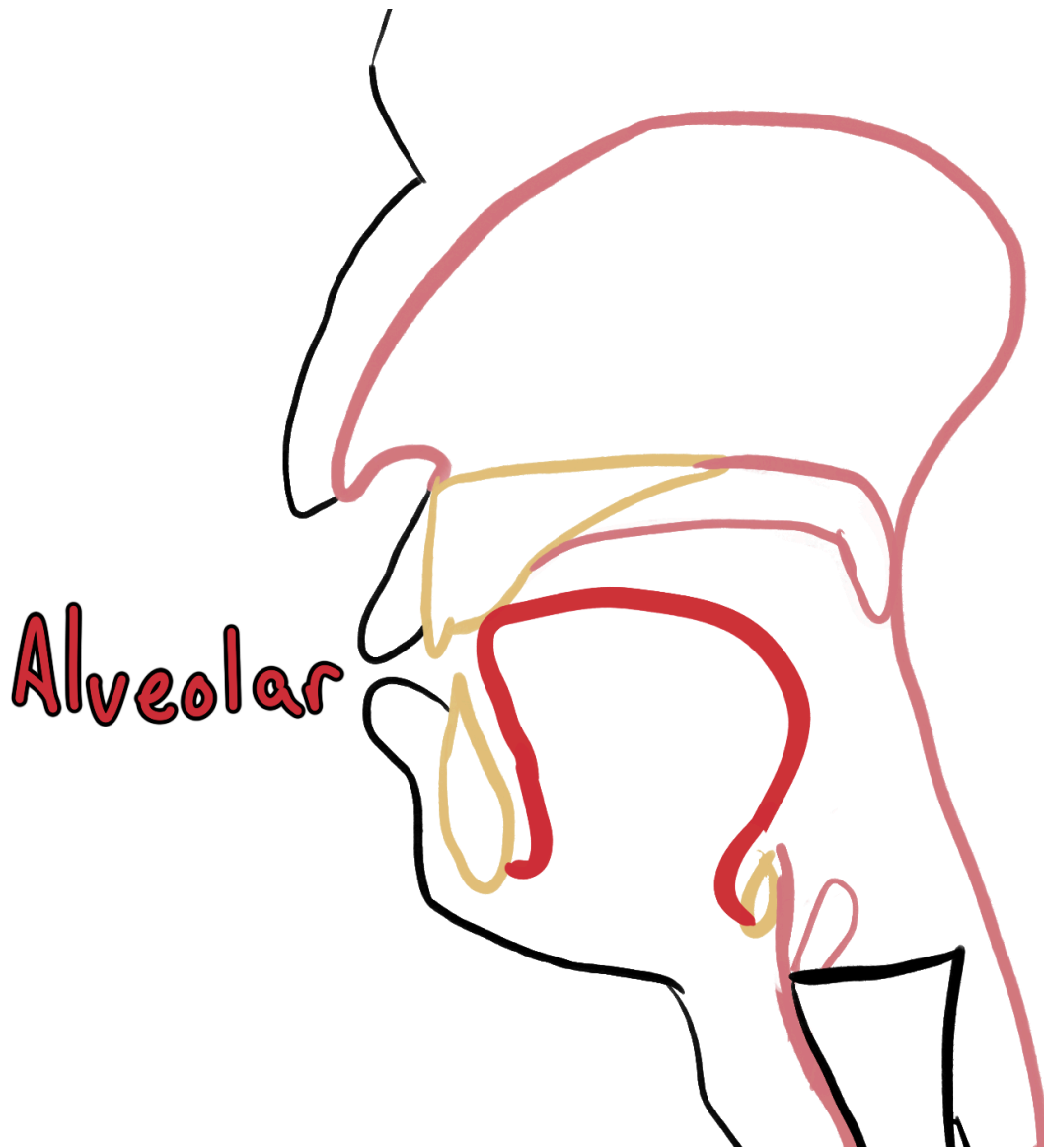
t d

n

s z



Place



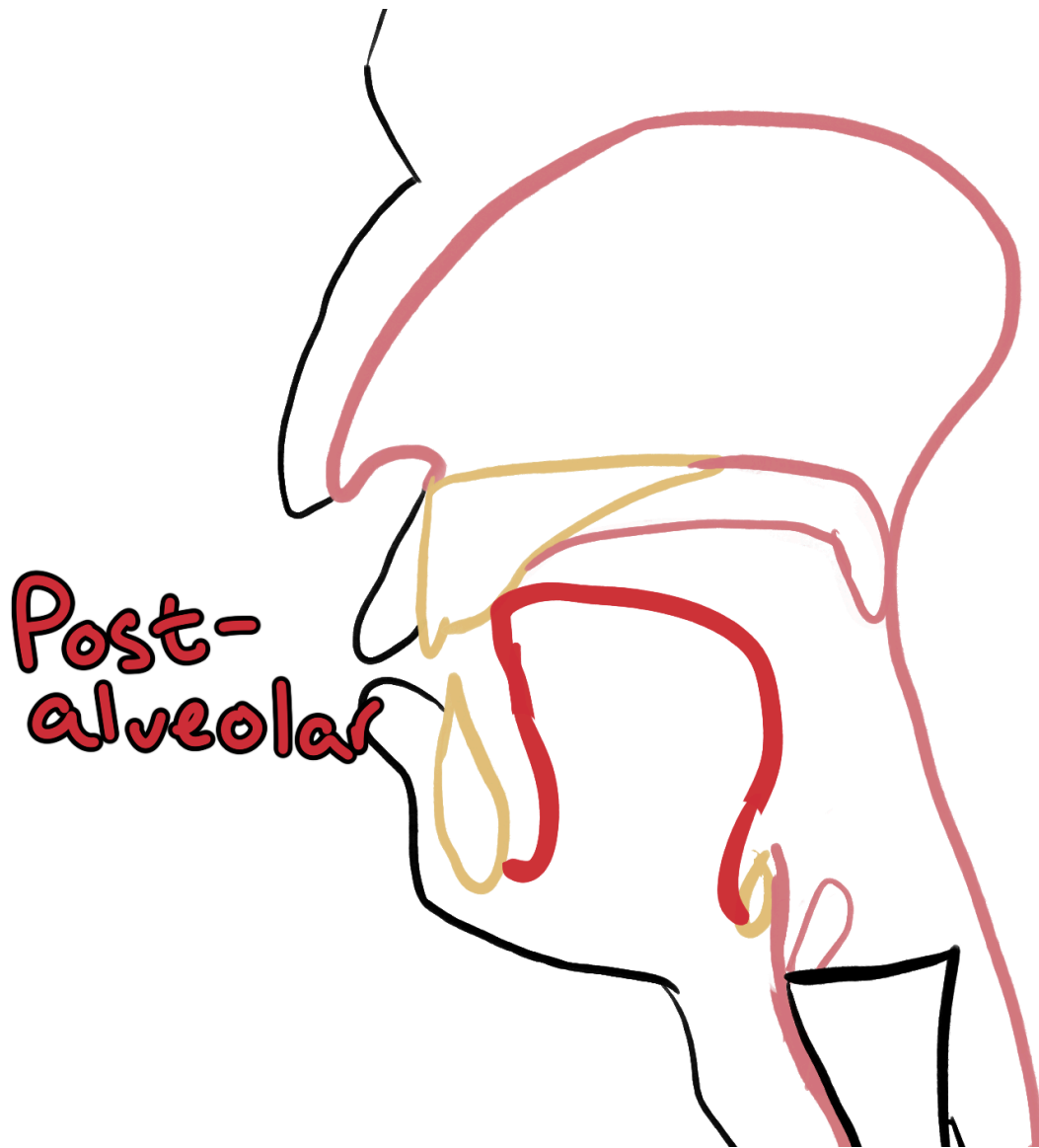


## Place

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |



Place





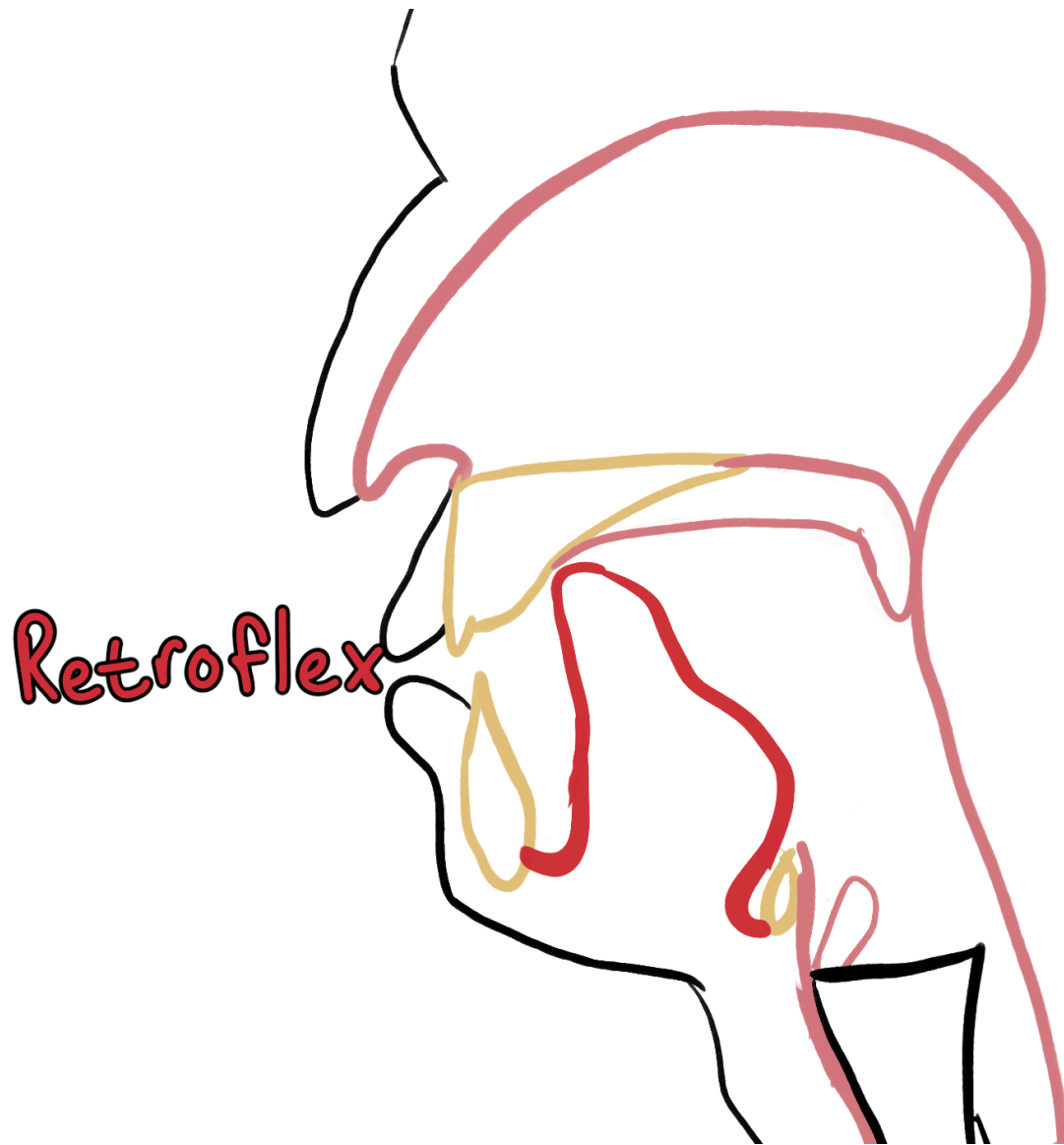


## Place

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
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| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |



Place



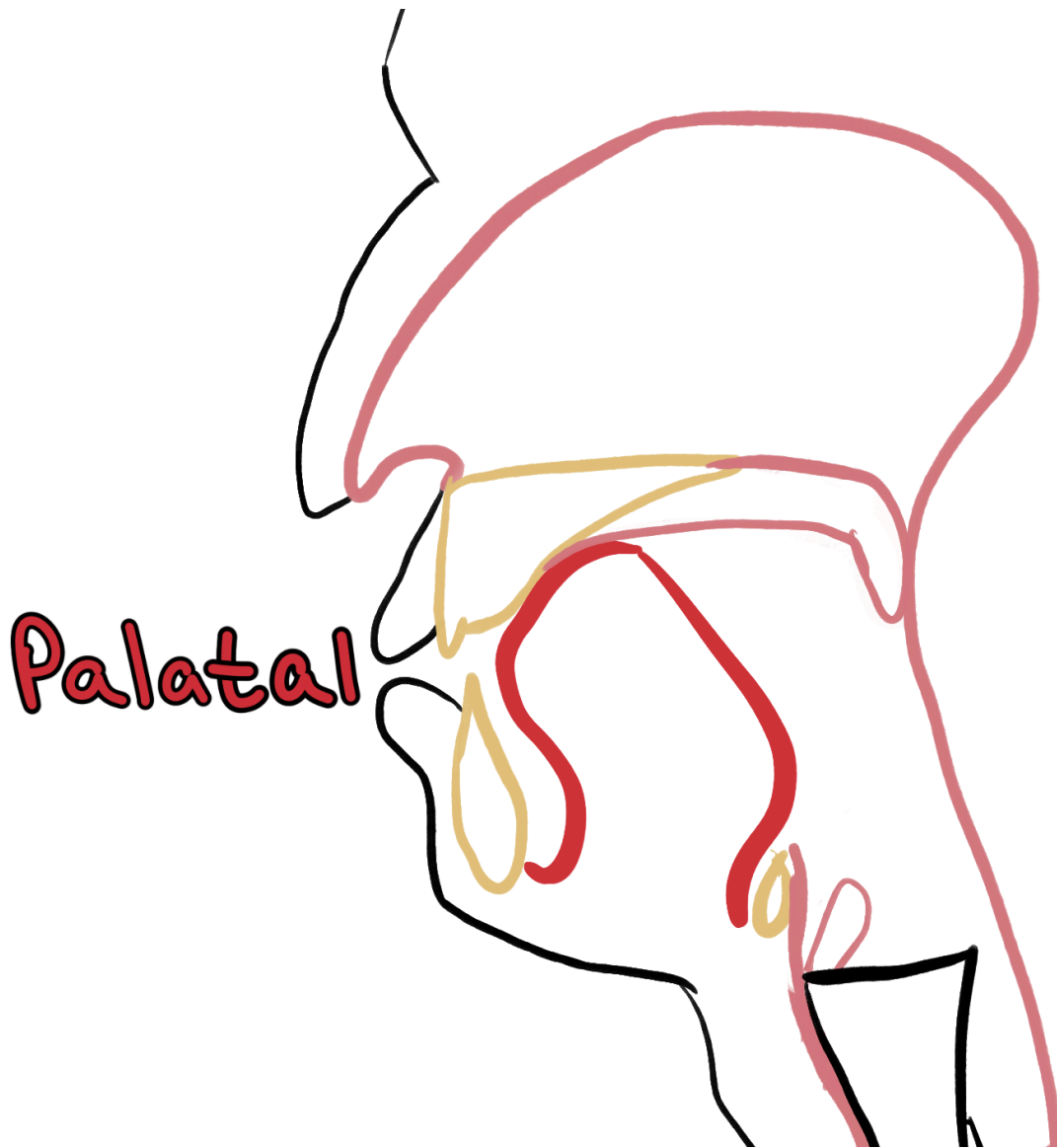


## Place

|                     | Bilabial |   | Labiodental |   | Dental |   | Alveolar |   | Postalveolar | Retroflex |   | Palatal |   | Velar |   | Uvular |   | Pharyngeal |   | Glottal |   |
|---------------------|----------|---|-------------|---|--------|---|----------|---|--------------|-----------|---|---------|---|-------|---|--------|---|------------|---|---------|---|
| Plosive             | p        | b |             |   |        |   | t        | d |              | ʈ         | ɖ | c       | ɟ | k     | g | q      | ɢ |            |   | ʔ       |   |
| Nasal               | m        | ɱ |             |   |        |   | n        |   |              | ɳ         |   | ɲ       |   | ŋ     |   | ɴ      |   |            |   |         |   |
| Trill               | ʙ        |   |             |   |        |   | r        |   |              |           |   |         |   |       |   | ʀ      |   |            |   |         |   |
| Tap or Flap         |          |   | v           |   |        |   | ɾ        |   |              | ɽ         |   |         |   |       |   |        |   |            |   |         |   |
| Fricative           | ɸ        | β | f           | v | θ      | ð | s        | z | ʃ            | ʒ         | ʂ | ʐ       | ç | ʝ     | x | χ      | ʁ | ħ          | ʕ | h       | ɦ |
| Lateral fricative   |          |   |             |   |        |   | ɬ        | ɮ |              |           |   |         |   |       |   |        |   |            |   |         |   |
| Approximant         | w        | ʋ |             |   |        |   | ɹ        |   |              | ɻ         |   | j       |   | ɰ     |   |        |   |            |   |         |   |
| Lateral approximant |          |   |             |   |        |   | l        |   |              | ɭ         |   | ʎ       |   | ʟ     |   |        |   |            |   |         |   |



Place



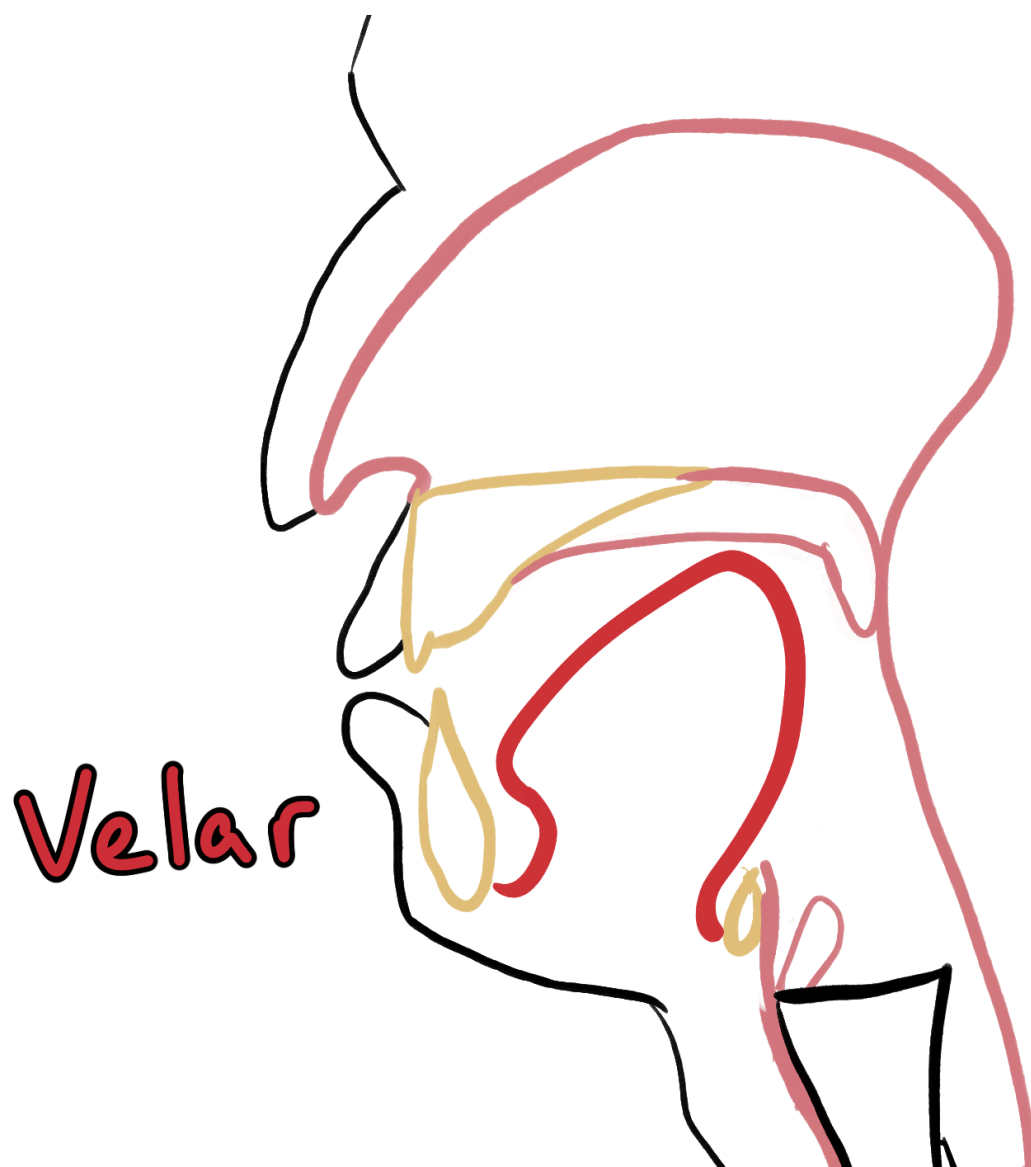


## Place

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|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | ɰ        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |



Place



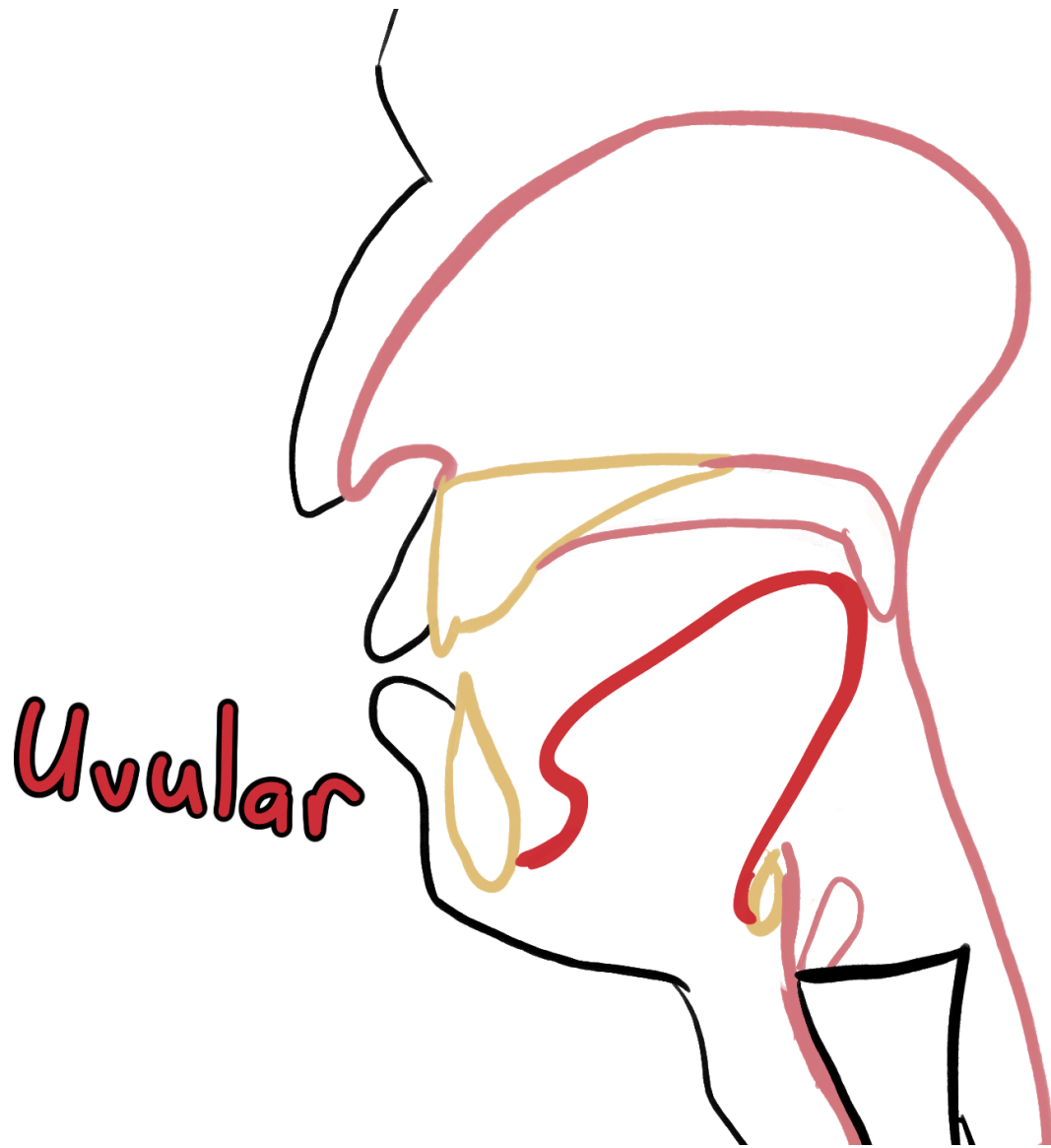


## Place

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|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |



Place



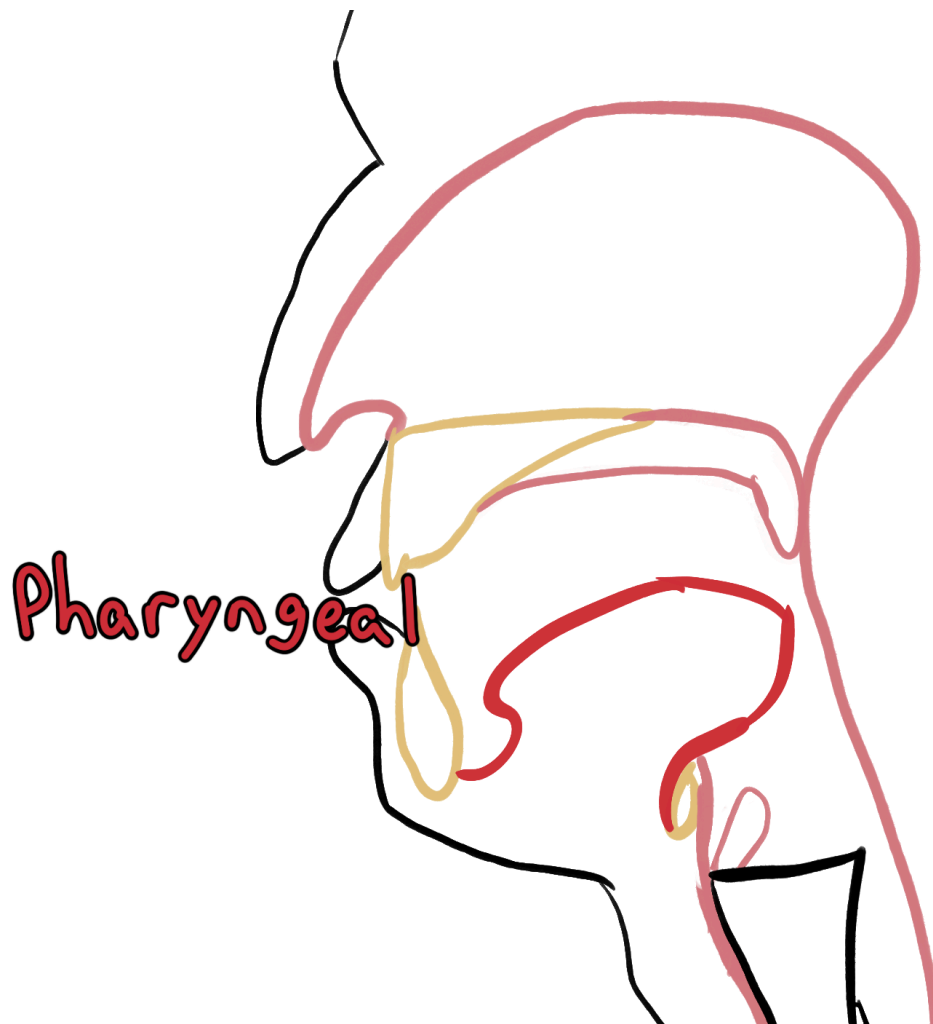




## Place

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|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | ʋ        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## Place





## Place

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
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| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## Place

There are also terms based on the active articulator involved.

- Front of tongue: Coronal
  - Tip of tongue: Apical
  - Blade of tongue: Laminal
- Back of tongue: Dorsal

## Place

| Apical      | Laminal   |
|-------------|-----------|
| ʂ, ʐ        | ç, ʝ      |
| (retroflex) | (palatal) |

## Manner

### Manner

What happens at each place of articulation.

### Manner

#### Oral Stops/Plosives

- Full constriction between articulators
- No oral airflow
- No nasal airflow



## Manner

### Oral Stops/Plosives

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ⱱ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## Manner

### Nasal Stops

- Full constriction between articulators
- No oral airflow
- Nasal airflow

## Manner

### Nasal Stops

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ⱱ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |



## Manner

### Trills & Taps

- Trills
  - Articulators held loosely together, allowing for airflow to generate the Bernoulli effect
- Taps
  - A very fast, glancing articulation.

## Manner

### Trills & Taps

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | ʀ        |              |           |         |       | ʁ      |            |         |
| Tap or Flap         |          | ⱱ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## Manner

Nasals and trills are assumed to be voiced, but if they're voiceless we use an under-dot to indicate that.

ᵰ ᵱ  
ɽ

## Manner

### Fricatives

- Close, but not complete constriction of articulators
- Turbulent airflow



## Manner

### Fricatives

|                     | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---------------------|----------|-------------|--------|----------|--------------|-----------|---------|-------|--------|------------|---------|
| Plosive             | p b      |             |        | t d      |              | ʈ ɖ       | c ɟ     | k ɡ   | q ɢ    |            | ʔ       |
| Nasal               | m        | ɱ           |        | n        |              | ɳ         | ɲ       | ŋ     | ɴ      |            |         |
| Trill               | ʙ        |             |        | r        |              |           |         |       | ʀ      |            |         |
| Tap or Flap         |          | ɸ           |        | ɾ        |              | ɽ         |         |       |        |            |         |
| Fricative           | ɸ β      | f v         | θ ð    | s z      | ʃ ʒ          | ʂ ʐ       | ç ʝ     | x ɣ   | χ ʁ    | ħ ʕ        | h ɦ     |
| Lateral fricative   |          |             |        | ɬ ɮ      |              |           |         |       |        |            |         |
| Approximant         | w        | ʋ           |        | ɹ        |              | ɻ         | j       | ɰ     |        |            |         |
| Lateral approximant |          |             |        | l        |              | ɭ         | ʎ       | ʟ     |        |            |         |

## Complex Consonants

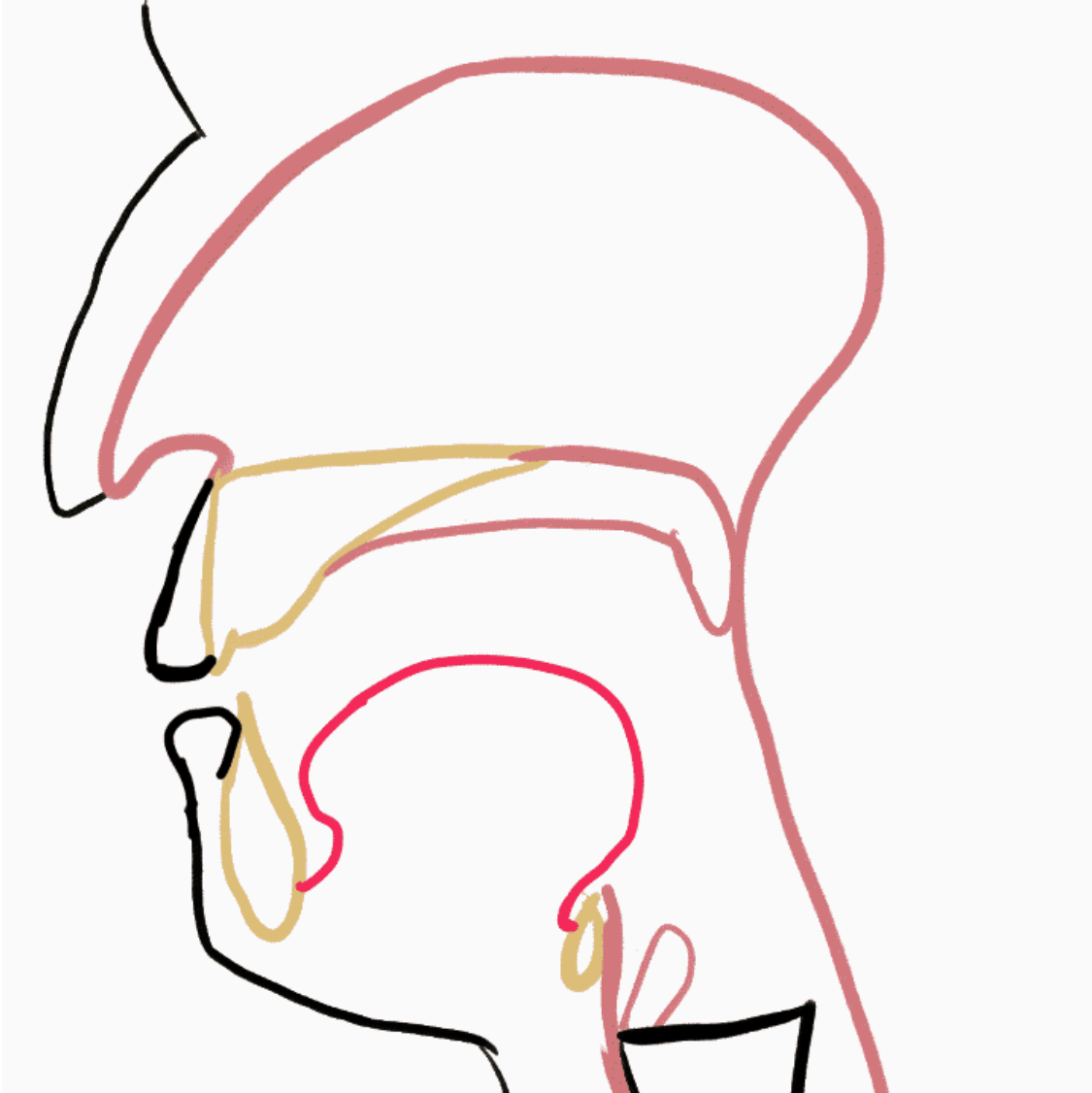
### Complex Consonants

- Doubly Articulated
- “Contour” segments



## Complex Consonants

Doubly Articulated



k



## Complex Consonants

Doubly Articulated

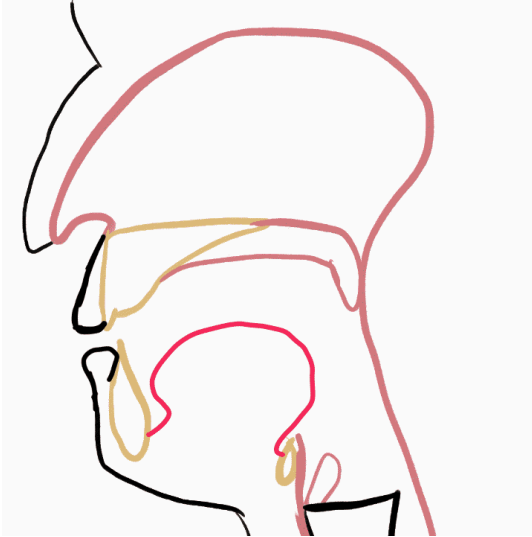


w



## Complex Consonants

### Contour Consonants



$\widehat{d}_3$

## Complex Consonants

### Contour Consonants



l

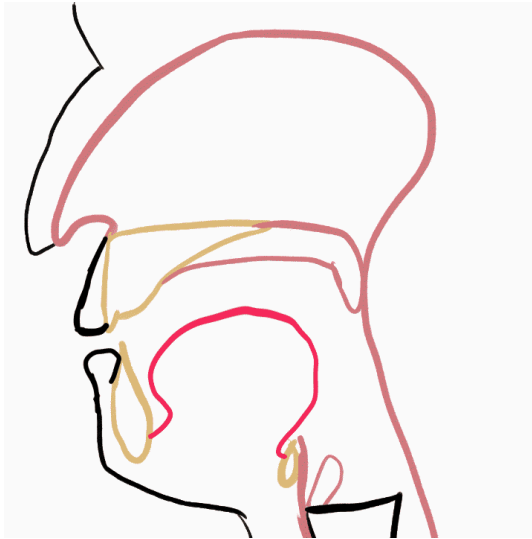
Sproat & Fujimura (1993)





## Complex consonants

### Contour Consonants

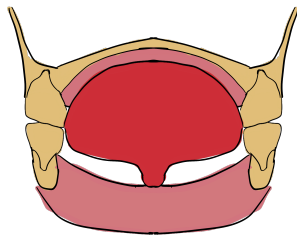


ɬ

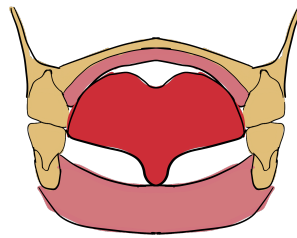
Sproat & Fujimura (1993)

## Other Classifications

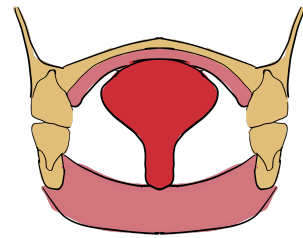
### Laterals



Stop closure



Fricative constriction



Lateral constriction

### “Rhotics”

Rhotics aren't a phonetically coherent set of sounds. Sounds classified as rhotics include

- alveolar trills
- uvular trills
- uvular fricatives



- velar fricatives
- alveolar taps
- retroflex approximants
- alveolar approximants
- whatever bunched [ɹ] is

## “Rhotics”

As the name would suggest, these are sounds spelled with characters derived from Greek “ρ”.

## References

### Bibliography

Delattre, P., & Freeman, D. C. (1968). A dialect study of American r's by x-ray motion picture. *Linguistics*, 6(44). <https://doi.org/10.1515/ling.1968.6.44.29>

Sproat, R., & Fujimura, O. (1993). Allophonic variation in English /l/ and its implications for phonetic implementation. *Journal of Phonetics*, 21, 291–311.